

SIMRAN KC

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PROFESSIONAL SUMMARY

AI Engineer and Researcher with 3+ years of experience in **production-grade NLP classifiers**, **RAG**, and **distributed inference pipelines**. Proven track record at **NASA IMPACT** in bridging the gap between research and software engineering to deploy scalable AI solutions in constrained environments.

SKILLS

Core AI & Research LLM Fine-tuning (LoRA, QLoRA), RAG, Agentic Architecture, Drift Detection and Continual Learning, Computer Vision, Reinforcement Learning

ML Frameworks PyTorch, Hugging Face (Transformers, PEFT), TensorFlow, Scikit-learn, OpenCV, Sentence Transformers, FastText, BERT

Data & Infrastructure Qdrant (Vector DB), OpenSearch, Redis, SQL Server, PostgreSQL, AWS (ECS, ECR, SQS, SNS, S3, EC2), Docker, GitHub Actions CI/CD

Software Engineering Python (FastAPI, Flask), JavaScript, WebSockets, OAuth 2.0, Pytest, Microservices

EXPERIENCE

Graduate Research Assistant, NASA IMPACT

Aug 2024 – Present

Huntsville, AL, USA

- Fine-tuned and evaluated **FastText** and **BERT-based classifiers** for multi-label classification of NASA Science Divisions, achieving **96% Precision** and **96% F1-score** and single label classification for relevancy tagging, achieving (**98% Recall**). Implemented a **Conformal Uncertainty-based alpha thresholding** mechanism for SDE relevancy tagging to establish a statistically rigorous human-in-the-loop workflow that **reduced manual curation by 95%**.
- Built a **distributed, horizontally scalable inference pipeline** using FastAPI, Redis, and worker processes, enabling dynamic model loading/unloading to serve multiple classifier models via a single API endpoint on resource-constrained infrastructure.
- **Co-authored technical abstracts** regarding ML-assisted data curation and NLP applications for the **NASA Marshall 65 Spring Showcase** and **AGU** (Full list: [Google Scholar](#)).
- Developed **data acquisition pipelines** for NASA's Deep Research Agent ([Accelerated Knowledge Discovery](#)) implementing custom resolvers and scrapers to support automated ingestion of technical and scientific documents.

Python AI Developer, Mutual Art

Sep 2023 – Jun 2024

London, England (Remote)

- Developed a **Vector-Based Recommendation System** processing 80k artworks weekly, utilized **Sentence Transformers (MiniLM)** for feature extraction and **Scikit-learn PCA** to retain 99.99% variance reducing storage by 90%, deploying a containerized **Flask API** on **AWS ECS** integrated with **Qdrant** and **OpenSearch**.
- Proposed and implemented a **Computer Vision Duplicate Detection Engine** using **OpenCV (ORB features)** and **RANSAC-based homography estimation** for scale-invariant matching; implemented a **Union-Find** algorithm in **Python** to cluster repeated sales, increasing platform conversion rates by 200%.
- Led the development of an **Automated Multimodal Description Service** by fine-tuning **IDEFICS-2 (8B)** via **LoRA (Rank-16)** and **4-bit NF4 quantization** to build a 24/7 inference pipeline on GPU servers using **AWS SQS/SNS**. Used **BLIP's Image-Text Matching** to validate and filter model hallucinations.

Associate Software Engineer, Sireto Technology

Sep 2022 – Sep 2023

Lalitpur, Nepal

- Developed **KuberIDE**, a microservices-based web IDE, implementing **OAuth** sessions and secure API key management for 500+ users to reduce environment setup time from hours to minutes.
- Engineered a high-performance **WebSocket proxy** to enable real-time **Haskell Language Server** IntelliSense with **<200ms latency**, driving a 300% increase in platform adoption.
- Optimized deployment workflows using **Docker multi-stage builds** (40% image size reduction) and **GitHub Actions CI/CD**, maintaining 80% code coverage via **Pytest**.

EDUCATION

Master of Science in Computer Science The University of Alabama in Huntsville <i>Relevant Coursework: Machine Learning, Advanced Operating Systems, Cloud Computing</i>	Aug 2024 – May 2026 GPA: 4.0/4.0
Bachelor of Computer Engineering Tribhuvan University, IOE Pulchowk Campus <i>Full Undergraduate Scholarship Recipient (Top 1% of the university)</i>	2017 – 2022 GPA: 3.7/4.0

PROJECTS

Production Agent Lifecycle Management for LLM Systems <i>Python, PyTorch, Hugging Face</i> MLOps framework for continuous reliable performance of LLM classifiers in production. Implements data drift detection using statistical tests, active learning strategies for sample selection, and incremental fine-tuning pipelines.
Pacman RL Simulation Engine Built a JavaScript Reinforcement Learning simulator with custom environment modeling, explicit state transitions, and an Approximate Q-Learning agent, deployed as a client-side web app on GitHub Pages.

AWARDS AND CERTIFICATIONS

Machine Learning in Production DeepLearning.AI (2025)
2nd Place Impact X-athon 2024 – NASA-sponsored innovation competition
Codepath Fellow (2024)
Microsoft Learn Student Ambassador Microsoft – Selected for Microsoft Summit from APAC region among 10,000+ applicants (2019 – 2021)
Machine Learning Coursera – Stanford Online (2019)